

can be considered for increased reliability and compactness. However, an ultra-efficient transformer will be needed to ensure high efficiency of the circuit.

Let us now take a look at high-efficiency rectifier circuits required to convert noise-like power to DC.

Rectification

The rectifier circuit is the heart of an energy harvesting circuit. It is responsible for converting the received energy from the transducer to DC voltage, sufficiently enough to feed battery-less electronics. Signal rectification exists in many forms including single-phase and three-phase, controlled (synchronous) and uncontrolled, and full-wave or half-wave. In most cases, non-synchronous, uncontrolled, and single-phase rectification is needed. Full-wave vs half-wave rectification is a decision that can be made by the designer based on losses, expected input voltage, and desired output voltage.

One common method of rectification is Schottky diode based voltage multiplier circuit, which not only rectifies the signal but also amplifies it. Schottky diodes present low-voltage switching, lower switching loss, and high efficiency, so they are preferred for low-power applications.

There are various topologies for voltage multipliers but among them Dickson voltage multiplier is widely used, as it offers a good trade-off between simplicity and performance. Fig. 3 depicts an n-stage Dickson rectifier. The diodes here can be implemented by MOSFETs, and the load resistance represents the input impedance of the circuit to be powered (microcontroller, sensor, etc).

The working of this rectifier is quite simple. Let us consider a single-stage rectifier consisting of D_1 , D_2 , C_1 , and C_2 . When the input voltage is negative, capacitor C_1 stores the charge through diode D_1 . When the input becomes positive, capacitance C_2 stores the charges resulting from the sum of the input voltage and the voltage across C_1 via D_2 . In case of ideal diodes, the voltage across C_2 without any load is given as:

$$2 \times (\sqrt{2}V_{RF, in} - V_{th})$$

The capacitors C_1 and C_2 can be implemented as stray capacitance

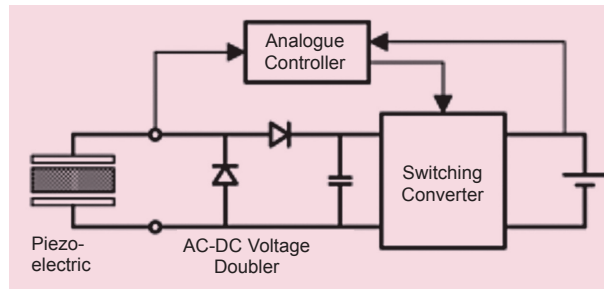


Fig. 4: An adaptive energy harvester circuit

on the chip, and the diodes as diode-connected MOSFETs that feature low 'on' resistance.

Adaptive power management

The output of the rectifier can be directly connected to the supercapacitor or the battery for charging. However, the variations of the piezoelectric voltage amplitude at the AC side of a rectifier, while the DC side voltage is retained fixed, can significantly reduce the amount of extracted power from the transducer. Various research suggest that based on the efficiencies achieved, the adaptive control circuits can enhance the efficiency of the rectifier circuit using maximum power point tracking like approach. They describe an adaptive energy harvester circuit from vibrational energy using a piezoelectric element as a transducer.

The system senses voltage from the transducer and at the output of the rectifier for controlling the output DC-DC converter that charges the battery. The aim of the analogue controller is to keep the output voltage of the rectifier and the open circuit voltage constant, when no load is applied.

Also, there might be multiple energy storage elements, one can be primary and the other secondary. Such an approach increases the reliability, because if the primary supercapacitor or battery is completely discharged, the system powers the load from the secondary storage device. Many applications ranging from low-power devices to electric vehicles use multiple supercapacitors/batteries for this kind of adaptive energy storage system.

Thus, if you are looking to implement an energy harvesting circuit on a chip, the above-mentioned circuits sum it up. However, you do not want a dedicated circuit on chip to realise the harvesting circuit in certain applica-

tions. There are specialised power management integrated chips (ICs) that integrate rectifiers, ultra-low power LDOs (low dropouts), and DC-DC converters for low-power energy for such applications and for those where integration of energy

harvesting circuits in a chip is not required. These ICs, moreover, feature low-power start-up because the power that is harvested is usually very small.

One such power management integrated circuit (PMIC) is e-peas's AEM40940 RF energy harvesting IC. It can start-up at RF input power from -19.5dBm to 10dBm. It features ultra-low-power boost converter, LDOs, and flexible energy management system that has selectable over-charge and over-discharge protection, suiting both rechargeable batteries and supercapacitors. The IC has provision for two energy storage devices, a primary and a secondary that automatically switch between each other when one of them is exhausted.

The adaptive control circuit involves open-circuit voltage sensing for maximum power point tracking (MPPT) every 0.33 second, and configurable MPPT. Impedance matching is also provided internally.

There are various other PMICs by e-peas intended for such energy-harvesting applications. S6AE10xA from Infineon is a solar-optimised energy harvesting solution and, likewise, there are many other such solutions for low-power management.

Conclusion

Battery-less solutions are flooding the market, and to support the energy harvesting technologies available today, there is an increasing demand for harvesting circuits that can not only make the power desirable for load but also maintain high efficiency while doing that. To achieve high efficiency, circuits like impedance matching and adaptive control systems are very critical. We have seen the PMICs available in the market that integrate all of the functionality. So, if integration on-chip is not needed, those can be preferred. **EFY**